

How Group Personality Composition Affects Person and Group Outcomes: An Integrative Analysis Using the Group Actor–Partner Interdependence Model

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A substantial part of people's social lives unfolds within groups. However, there is a notable research gap concerning if and how the personality characteristics that people bring to group interactions combine to predict person and group outcomes. In this study, we used the group actor–partner interdependence model (Kenny & Garcia, 2012) as a framework to integrate prior approaches and understand how the composition of two socially relevant personality traits—agency and communion—affects people and groups. We analyzed data from 432 participants ($M_{\text{age}} = 26.61$, 51% female) who formed 108 four-person groups and engaged in four different group tasks. Our findings yield three key insights: (a) At the person level, people's own trait levels were the main drivers of their behaviors, experiences, and performance. (b) At the group level, personality composition affected different outcomes than at the person level, with agency playing an overall more important role for group behaviors and experiences. (c) Notable composition effects at both levels emerged for conflict behavior: People who were similar to their group in terms of agency were more engaged in conflicts, and groups whose members had similar agency levels were more likely to experience conflicts as a whole. We contextualize our findings within a theoretical framework to better understand when and how personality composition in social interactions is important, and we review methodologies to capture its multifaceted components.

Keywords: group interactions, personality composition, agency, communion, group actor–partner interdependence model

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A substantial part of people's social lives unfolds within groups, from working with colleagues on projects, joining a book club, to meeting new neighbors in the community. These group interactions not only play a crucial role in shaping social relationships but also significantly impact performance in academic and work settings (e.g., Badura et al., 2022; Emery et al., 2013; Garcia et al., 2015; Laursen, 2005; Levine & Moreland, 1994). One key ingredient in group interactions is the way each interaction partner tends to think, feel, and behave toward others, which is captured in their stable personality traits (Roberts et al., 2006). In addition to each interaction partner's personality, a fundamental psychological assumption is that the composition of different personalities that interaction partners express shapes social processes, experiences, and outcomes

(Back et al., 2011; Hopwood et al., 2021; Wiggins, 1979). Accordingly, personality composition research targets a range of important questions in social contexts. Focusing on individuals within groups: Do the same people have different experiences depending on who they interact with in a group? Focusing on groups as a whole: Does the degree to which group members share similar personality traits shape the group's climate or joint performance?

Although the theoretical importance of personality composition effects on persons and groups has been widely recognized, our understanding of these effects has remained fragmented for several reasons. First, existing studies vary considerably in how they operationalize and investigate personality composition, with many focusing primarily on dyadic rather than group contexts. Second, the focus of most research is

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either on person-level outcomes (e.g., individual behaviors or perceptions) or group-level outcomes (e.g., group behavior and performance), with little integration between these perspectives (Moreland, 2010). Third, many studies examine isolated outcomes, such as individual satisfaction or performance on a single task. This narrow focus leaves open questions about the context-dependent nature of composition effects because it is unclear if findings generalize beyond the specific settings in which they were observed.

The present study aimed to address these challenges in three ways. First, we integrate theoretical perspectives and empirical insights from different psychological disciplines related to personality composition in social contexts, with a particular focus on group settings. Second, we introduce the group actor–partner interdependence model (GAPIM; Kenny & Garcia, 2012) as a conceptual and statistical framework to unify various operationalizations of personality composition effects in newly formed groups. Third, we apply the GAPIM analysis to a large data set containing 432 participants in 108 groups, where each group engaged in four different group tasks. Spanning a diverse range of person- and group-level outcomes, to our knowledge, this reflects the most comprehensive investigation of personality composition effects in groups to date.

Theoretical Perspectives on Personality Composition in Social Contexts

Grounded in an interactionist perspective that views people's experiences (i.e., behaviors and perceptions) as outcomes of an interplay between individual and contextual characteristics (Lewin, 1936; Mischel & Shoda, 1995; Rauthmann, 2021; Reynolds et al., 2010), a central element of any social interaction is the people present and the psychological traits they bring to the situation (Clark et al., 2020). To illustrate, Maja's experiences during social interactions may differ depending on whom she interacts with: She might behave and perceive others differently depending on how similar or different her personality is to those of her interaction partners, which may also affect how well they work together as a group (Bell, 2007; Cuperman & Ickes, 2009; Kristof-Brown et al., 2005; Overall et al., 2023).¹ As a result, each and every social interaction of Maja is likely shaped by the unique composition of personalities of all participants: her own (denoted as the actor personality), those of her interaction partners (partners' personalities), and a combination of both (Kenny, 2018).

The current work investigated the role of personality composition in group interactions within the framework of interpersonal theory (e.g., Carson, 1969; Leary, 1957; Sullivan, 1953). Interpersonal theory describes personality along two fundamental dimensions that capture people's functioning and behaviors in social contexts: agency, a trait linked to an individual's drive to get ahead in social situations, including attributes such as dominance and assertiveness, and communion, a trait linked to the goal to get along with others, including attributes such as warmth and empathy (Kiesler, 1983; Roche & Cain, 2021; Sullivan, 1953; Wiggins, 1979). We focus on agency and communion because of their cross-disciplinary relevance. Extending from social psychology to leadership, clinical, or educational research (e.g., Abele-Brehm & Wojciszke, 2019; Koenig et al., 2011; Pennings et al., 2018; Rau et al., 2023), both dimensions are widely used to understand people and groups.²

Additionally, interpersonal theory features specific assumptions on how the composition of interaction partners' agency and communion

manifest in social settings (Kurzius et al., 2022; Sadler et al., 2010). These assumptions connect to two broad theoretical perspectives that have fueled research on personality composition: person–environment fit perspectives (Kristof-Brown & Guay, 2011; van Vianen, 2018) and similarity-attraction theory (Byrne, 1971). Person–environment fit perspectives propose two composition patterns (Cable & Edwards, 2004; Muchinsky & Monahan, 1987): complementary fit, where a diversity among person and group attributes leads to positive outcomes (i.e., a person's traits are complemented by the group and vice versa), and supplementary fit, assuming that similar characteristics among person and the group are beneficial (in line with the similarity hypothesis).

These two patterns resonate with interpersonal theory (Carson, 1969; Sadler et al., 2010), where agency is assumed to evoke submissive responses in others (complementary pattern), while communion elicits similar responses (supplementary pattern). The rationale behind the positive effects of these two distinct patterns is that the intrinsic motives tied to agency (to get ahead) and communion (to get along) are met when complementarity occurs (Cable & Edwards, 2004; Horowitz et al., 2006). For instance, if Maja's high agency leads her to seek control and make decisions, she benefits most when other group members are less agentic and allow her to take charge. Conversely, her goals could go unmet if others in the group also strive for leadership. The assumption that a diversity in agency plays a role in building and keeping a social hierarchy within groups is also echoed by leadership emergence theories (Van Vugt, 2006). By contrast, the desire to foster harmonious relationships found in individuals with high levels of communion is most effectively met when all members of one's group have a similar desire.

Similarity-focused research assumes that a match between a person and their environment generally leads to positive outcomes in a supplementary manner. For example, we would expect that Maja's interactions are more positive when her own agency and communion levels match those of her interaction partners. In group settings, organizational research has debated the merits of similarity versus diversity of group member's personality: while similarity perspectives suggest that less similar groups face greater risk of conflict, potentially undermining group climate and performance (e.g., Halfhill et al., 2008), diversity advocates highlight the benefits of having varied opinions in the group (Moreland, 2013; van Knippenberg et al., 2007).

To summarize, person–environment fit and similarity perspectives are well-established theoretical frameworks that propose partially contrasting assumptions concerning the benefits of different composition patterns for agency and communion. In the following sections, we review empirical research on the composition effects of agency and communion in social contexts. Specifically, we focus on (a) the operationalization of composition in social interactions and (b) the outcomes considered in different research fields.

¹ Personality traits are conceptualized as latent constructs expressed through behavioral and perceptual tendencies, which, in combination with contextual factors, predict behaviors and perceptions in social settings (Back et al., 2011).

² Different labels are used across research domains (e.g., competence vs. warmth, plasticity vs. stability, power vs. intimacy; Abele-Brehm & Wojciszke, 2019) and focus on different elements of agency and communion, such as personality (Gebauer et al., 2012), interaction behavior (Kurzius et al., 2022), motives (Horowitz et al., 2006), or interpersonal judgments (Schauf et al., 2023). Despite these differences, the underlying frameworks share considerable conceptual overlap.

Empirical Findings on Personality Composition in Social Contexts

Within the fields of personality, organizational, and educational research, personality composition effects have been investigated from different vantage points. On the one hand, person-centered approaches consider composition in terms of how the actor's personality traits (e.g., Maja) relate to those of interaction partners and shape the actor's behaviors or experiences. On the other hand, group-centered approaches treat groups as collective units and investigate how personality at the group level relates to outcomes like the group's overall performance. Hence, these two perspectives differ in how they operationalize and measure personality composition effects—considering them as person or group attributes—and in their focus on either person-level or group-level outcomes.

Person-Centered Approaches

Much of the empirical work on personality composition has centered on how it affects individuals in dyadic relationships. In dyadic contexts, composition is usually operationalized through three components: the actor's personality trait (e.g., Maja's agency level), the partner's trait (e.g., Tim's agency level), and the interaction between actor and partner personality. This approach to studying composition is best captured in the established actor-partner interdependence model (APIM; Cook & Kenny, 2005). In the APIM, outcomes are measured at the person level, that is, actors and partners provide individual ratings.

Focusing on self-perceptions, findings show that extraversion and agreeableness—the Big Five dimensions that are most closely associated with agency and communion (Entringer et al., 2022)—have consistent effects on how people perceive their social relationships (Humberg et al., 2023; Körner & Altmann, 2023; Weidmann et al., 2017). That is, people with high extraversion and agreeableness generally report more satisfaction with their relationships (actor effect), while people seem to be more attracted to extraverted partners and satisfied with more agreeable partners (partner effect). Apart from these main effects, interaction effects between actor and partner agency or communion are typically found to be negligible for how people perceive their relationships.

By contrast, interactive composition effects seem to play a more important role in nonromantic contexts. Specifically, people tend to like unfamiliar others who have similar levels of extraversion (agentic domain) during initial interactions (Cuperman & Ickes, 2009), while similarity in agreeableness (communion domain) seems to be beneficial only at high levels. That is, interactions are rated as more successful—based on people's perceptions of interaction quality—when both score high on agreeableness but worsen when both score low (Arellano-Véliz et al., 2024; Cuperman & Ickes, 2009). Thus, the findings from dyadic research suggest that the composition of agency and communion can shape initial social interactions by predicting person-level behaviors and perceptions, though they do not clearly support either similarity or fit perspectives. Applying these insights to group interactions is, however, challenging: Multiple interaction partners introduce greater complexity in interaction dynamics (Moreland, 2010) and may require different operationalizations of composition.

To address the issue of operationalization, several studies have adapted the logic behind the dyadic APIM predictors (i.e., Actor,

Partner, Actor $\times$ Partner interaction) to group settings, assessing composition as the actor's personality trait level, the group's (average) personality, and the interaction between actor and group personality (e.g., Kristof-Brown et al., 2005; Schmidt et al., 2012). Accordingly, this approach focuses on the individual similarity between an actor and their group, essentially measuring the actor-group fit. Another approach considers composition in terms of variability of a personality trait within the group using dispersion measures such as the standard deviation or variance as predictors (e.g., Gonzalez-Mulé et al., 2014). In this case, composition is viewed as the overall similarity among group members, referred to as group similarity. The assumption behind the group similarity metric is that high similarity (i.e., low variability) in a personality trait indicates a supplementary fit, while low similarity (i.e., high variability) reflects a complementary fit. In only a few instances, researchers have considered both actor-group fit and group similarity measures to investigate personality composition from different angles (Peeters et al., 2006; Prewett et al., 2018).

Given the different approaches to assessing composition effects, it is not surprising that research findings are challenging to interpret, compare, and integrate. Regarding agency, several studies suggest that a complementary pattern improves people's experiences in groups. For example, a complementary actor-group fit (e.g., Maja's agency level is substantially different from that of her group) has been linked to more cooperative behavior (Schmidt et al., 2012). Additionally, lower group similarity (e.g., people in Maja's group are diverse in terms of agency) has been linked to fewer counterproductive behaviors and better individual performance (Gonzalez-Mulé et al., 2014; Prewett et al., 2018). However, other studies have produced rather inconsistent findings. An example of this is how agency composition relates to a person's perceived satisfaction with their team: While one study suggests that people with a complementary actor-group fit in agency are more satisfied with their group (Kristof-Brown et al., 2005), another reports that people in groups with low agency similarity are less satisfied (French & Kottke, 2013). Then again, results suggest that such diversity reduces satisfaction only for actors scoring low on agentic traits (Peeters et al., 2006). Thus, there is some support for the idea that complementary agency effects are beneficial in groups; however, the robustness of these findings is called into question by variations in how group composition is defined and the diversity of outcomes assessed across studies.

Research on communion composition in groups is even scarcer. Some evidence suggests that people behave and perceive others more positively when group members share similar communion levels (i.e., high group similarity; Gonzalez-Mulé et al., 2014; Tekleab & Quigley, 2014). However, other research indicates that only the actor's own level of communion is meaningfully linked to their perceived satisfaction with the group or their own performance (Peeters et al., 2006; Prewett et al., 2018), which could suggest that beyond the actor's communion, the composition in the group is less relevant for these outcomes. Beyond these self-rated personal experiences, little research has investigated how communion composition may shape people's behaviors or performance in groups.

In sum, although some evidence leans toward the fit perspective suggesting that complementary agency and supplementary communion levels in groups could be beneficial, drawing firm conclusions is difficult. Key issues include significant variation in the measurement of group composition and the outcomes analyzed across studies. Furthermore, many studies depend on self-reported outcome measures, and sample sizes tend to be small because

collecting data from groups is often more challenging than from individuals. To our knowledge, no research has yet comprehensively explored how different measures of personality composition (e.g., actor-group fit and group similarity) relate to different person-level outcomes. The present study addresses these gaps, offering new insights into the role of personality composition in a wide range of people's behaviors, perceptions, and individual performance in group interactions.

Group-Centered Approaches

Researchers in work and educational environments are interested in not only how personality composition shapes individual experiences but also how it affects the outcomes of groups as a whole, such as a team's joint performance or their work climate (Bell, 2007; Halfhill et al., 2008; LePine et al., 2011; Moreland et al., 1996). A debate similar to that of in-person-centered approaches continues regarding the best way to operationalize personality composition for predicting group outcomes. The two most common methods are aggregating the personality scores of all group members to a group personality or, similar to the person level, using the variance across personality scores to investigate group similarity.³

Most group-centered research on personality composition has focused on how it relates to group performance (e.g., Bell, 2007; Halfhill et al., 2008; Kramer et al., 2014; Mohammed & Angell, 2003). Meta-analytic findings focusing on group performance outcomes suggest that (a) groups with higher agency averages (i.e., stronger group personality) and greater agency diversity (i.e., lower group similarity) perform better on tasks, though these effects are generally small, and (b) groups with a higher average and less diversity in communion tend to achieve better performance (Prewett et al., 2009).

While these overall patterns align with fit perspectives, two critical points highlight the need for more research on personality composition at the group level. First, results vary considerably depending on the measurement of performance (e.g., objective criteria, external observer, aggregated group member ratings) and the nature of the task (e.g., group discussions, problem solving, creative tasks; cf. Halfhill et al., 2008; Kramer et al., 2014; Mohammed & Angell, 2003; Prewett et al., 2009). Second, few studies have investigated the link between composition and group outcomes beyond performance. One specific composition-outcome link that researchers have been interested in is between a group's diversity (i.e., low group similarity) and group conflict, which is connected to a group's overall climate and effectiveness (De Dreu & Weingart, 2003). Initial results suggest that low agency similarity (i.e., high diversity) does not account for increased group conflict (Mohammed & Angell, 2004), while increased diversity in communal traits could contribute to more conflict in groups (Lee & Park, 2020).

What insights can we gain from the fragmented literature on personality composition across person- and group-centered approaches? We identify three key points. First, personality composition beyond the actor's own agency and communion levels can shape an individual's behaviors, experiences, and performance within group interactions as well as group-level outcomes like performance. However, the relative importance of certain composition components at both levels remains unclear. Second, because different composition operationalizations produce different results at person and group levels, it makes sense to assess several indicators of composition and systematically study their

effects across person and group levels. Third, composition effects may not only vary by indicator but probably also differ across outcome domains (e.g., behaviors, perceptions, performance) and assessment methods (e.g., group member rating, observer ratings, objective criteria). Thus, an integrative approach that combines composition measures and includes diverse outcomes is important for advancing our understanding of how personality composition shapes people's experiences in group interactions.

An Integrative Framework: The GAPIM

In our research, we use the GAPIM developed by Kenny and Garcia (2012) as a conceptual and statistical framework for understanding agency and communion composition in group interactions. The GAPIM differs from traditional approaches by focusing on the relationships between group members rather than viewing composition as a single measure for a person or the entire group (Bonito, 2022; Garcia et al., 2015). Focusing on personality, this means that the model considers how each person's personality trait in the group relates to and interacts with every other group member's personality trait. Importantly, the GAPIM offers a flexible configuration based on the level of the outcome (i.e., individual, dyad, group). The present study centers on personality composition at person and group levels using the GAPIM framework, consistent with the prevalent focus in personality, educational, and organizational research. Exploratorily, we extend this focus to the dyadic level of personality composition within groups—an aspect that has been largely overlooked in existing literature.

GAPIM Configurations at the Person Level

The GAPIM-I ("I" indicating individual) focuses on person-level outcomes and extends the traditional APIM (Cook & Kenny, 2005) used in dyadic contexts. This version of the GAPIM integrates prior person-centered approaches by breaking down composition into four effects (see Figure 1). The first two effects include the actor's personality trait level (actor effect) and the average trait level of all group members except the actor (group[−1] effect). Thus, rather than portraying the group's overall average, the group(−1) effect reflects a distinct value for each person in the group.

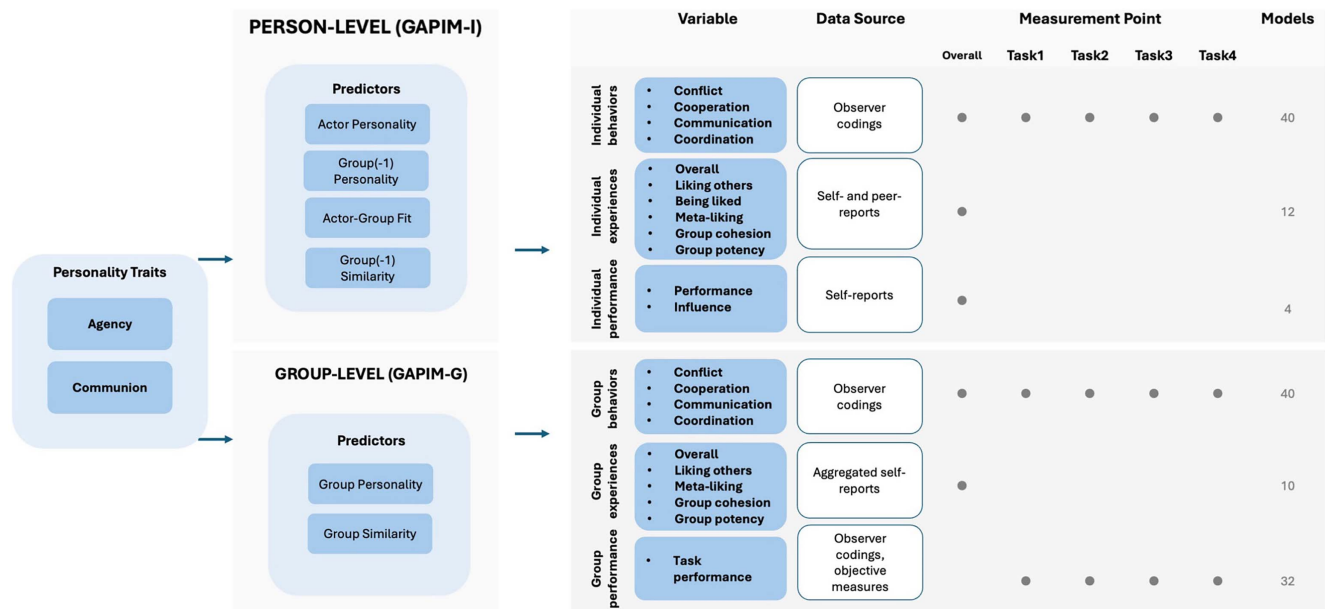
Additionally, while the dyadic APIM includes one interaction term between actor and partner personality, the GAPIM expands on this by including two composition components: the actor-group fit, reflecting the similarity between the actor and other group members, and group(−1) similarity, representing the average similarity among all possible pairs of interaction partners within the group (excluding the actor; Garcia et al., 2015). Unlike previous approaches that measured group similarity using a single metric (e.g., the standard deviation of a personality trait within a group; Gonzalez-Mulé et al., 2014), the concepts of actor-group fit and partner similarity offer a more nuanced perspective (Kenny & Garcia, 2012).

To illustrate how the GAPIM-I operates, consider Maja again, who scores high on agency (she is assertive and energetic). During a

³ Some researchers also consider minimum and maximum personality trait levels within a group to account for floor and ceiling effects (e.g., assuming that the person with the lowest or highest agency level determines the group's performance; Kramer et al., 2014; O'Neill & Allen, 2011; Prewett et al., 2009).

Figure 1

Conceptual Framework Showing the GAPIM Composition Components and Operationalization of Outcomes at Person and Group Levels



Note. Actor personality = individual level of agency or communion. Group(–1) personality = averaged levels of agency or communion in the group, excluding the actor. Actor-group fit = interaction between actor and group(–1) personality trait levels. Group(–1) similarity = averaged interaction among all possible pairs of partners' personality trait levels, excluding the actor. Group personality = averaged actor effects. Group similarity = averaged group(–1) similarity. For each listed outcome variable, one model with agency and one with communion as a personality composition variable were estimated, resulting in two models per outcome. Exploratory analyses at the dyadic level for liking and metaliking are presented in [Supplemental Material](#). GAPIM = group actor-partner interdependence model. See the online article for the color version of this figure.

group interaction, she finds herself in several conflicts with others about how to solve a group task. The four effects in the model show to what degree Maja's conflict behavior during the interaction is a function of (a) her own high agency level, (b) the average agency level of her interaction partners, (c) the degree to which she is more (or less) similar to her partners' agency, and (d) how similar (or diverse) her partners are in terms of agency. Drawing from the theoretical assumptions of the fit perspective and previous findings, Maja likely gets into more conflicts in groups wherein others also want to take charge due to high agency levels. On the other hand, a more diverse group in terms of agency could help reduce tensions, as people tend to fall into leader–follower roles. In this case, the GAPIM-I would show a significant negative effect of group(–1) similarity of agency.

GAPIM Configurations at the Group Level

The GAPIM-G (“G” indicating group) is tailored to group-level outcomes and includes the two most frequently used composition metrics from group-centered approaches: group personality, which captures the average personality traits of all group members, and group similarity, which represents the average partner similarity among all group members (Figure 1). Thus, the GAPIM-G focuses exclusively on group-level predictors. The two composition measures provide insights into how group members' personality traits either combine additively or interactively (i.e., whether the group is more than just the sum of its members) to impact group outcomes (Kenny & Garcia, 2012; Levine & Moreland, 1994). Accordingly, the model

provides an integrative framework to examine, for example, how agency and communion composition is linked to how Maja's group performs on a joint task.

Considering a Special Case: GAPIM Configurations at the Dyad Level

An overlooked aspect in research is how personality composition affects dyadic processes and outcomes among group members. Consider Maja's liking perceptions: knowing both her overall group preference and her specific liking of Tim, Sarah, or Alex allows us to disentangle whether she likes certain partners more when they share similar personality trait levels. Conceptually, this is important because assessments at different levels (e.g., liking the group as a whole vs. liking others individually) could be driven by different underlying processes that do not necessarily depend on each other (Moreland, 2010).

While the GAPIM framework can be adapted to study personality composition effects on dyadic outcomes (GAPIM-D; Kenny & Garcia, 2012), implementing such dyadic-level analyses poses several challenges. First, they require specific data in the form of directed dyadic ratings (Wong et al., 2024), which are typically collected via round-robin designs wherein actors rate all interaction partners and vice versa. Second, incorporating the dyadic level significantly increases model complexity due to dependencies in the data between actors, partners, and remaining group members. While a comprehensive exploration of the GAPIM-D is beyond this

article's scope, we highlight its conceptual importance and provide an illustrative application in the [Supplemental Materials](#).

The Present Study

The overarching goal of the present research is to unify existing approaches to studying personality composition in groups in three important aspects. First, in terms of trait content, we aim to unify existing research by focusing on two essential interpersonal traits—agency and communion—to examine if and how their composition plays a role in group interactions. We test varying theoretical assumptions on whether complementary (diverse) or supplementary (similar) compositions of agency and communion in groups can be beneficial for both individual experiences and group phenomena. To do so, we study agency and communion composition in initial interactions among previously unacquainted small groups ($N = 408$ in 108 groups).

Second, to unify prior modeling approaches, we use the GAPIM (Kenny & Garcia, 2012) as an integrative framework for simultaneously modeling various indicators of composition at different levels. At the person level, we examine how composition components such as actor and group effects, group-actor fit, and group similarity affect individual outcomes. At the group level, we focus on the associations between overall group personality and group similarity and group outcomes. As an exploratory extension, we demonstrate how the GAPIM framework can be adapted to analyze personality composition effects at the dyadic level.

Third, we unify different outcome domains and data sources by adopting a multimethod approach. This inclusive strategy considers a broad spectrum of outcomes at the person and group level. That is, we consider observer-coded behaviors at both levels, subjective perceptions of all interaction partners, and different indicators of group performance across four different tasks (i.e., designing a poster, a survival on the moon mission, building a bridge with small plastic bricks, and discussing a moral dilemma) characterized by varying task demands (i.e., creativity, generating arguments).

To our knowledge, this study is the first to use the GAPIM framework to investigate personality composition in group interactions. Given the multiple operationalizations of personality composition, the inconsistent results of previous studies, and the fact that the GAPIM has not yet been empirically tested with continuous variables such as personality traits, it is difficult to predict specific effects. Therefore, we did not formulate directional hypotheses on how the different composition components might predict the different outcomes. However, we preregistered our research questions, key constructs, and modeling approaches to ensure the transparency of our research. We focus our investigation on group interactions in the early stages of the acquaintance process for two reasons: First, initial interactions are the basis for subsequent relationships and therefore provide an important starting point for understanding personality effects in group dynamics. Second, studying initial interactions offers the valuable opportunity to analyze the effects of interaction partners' agency and communion devoid of already existing relationship dynamics within the group (cf. Ambady & Skowronski, 2008). For a clear presentation of our research agenda, [Figure 1](#) summarizes the main outcomes of interest at person and group levels, as well as the number of models that were analyzed. The online [Supplemental Materials](#) contain information on the exploratory dyadic analyses.

Method

Transparency and Openness

We describe our sample size determination, any data exclusions, and all measures used in the study. The present research involves secondary data analyses of data from a multimethod study focusing on interpersonal attraction and group performance in small group interactions.⁴ Ethical approval for the study was granted by the ethics committee of the University of Leipzig (title: "Implicit Interpersonal Attraction in Small Groups," Protocol Number: SK 102014). The data have been analyzed in previous studies (Nikoleizig et al., 2019; Rau et al., 2020, 2022), none of them focusing on group composition effects. In the current article, we describe those procedures that are directly relevant to this study. Our preregistration of the research questions and the analysis plan is available on the Open Science Framework, along with the data, analysis code, and [Supplemental Materials](#) (Bleckmann et al., 2025; <https://osf.io/muaq5/>). [Supplemental Table OS1](#) provides an overview of all deviations from the preregistration. No special or custom-made materials were used in this study. All analyses were conducted in R (R Core Team, 2020), using the stats and lme4 packages (Bates et al., 2015); graphs were constructed using sjPlot (Lüdtke, 2013). Researchers interested in reproducing our results can use the available analysis code and output files.

Procedure

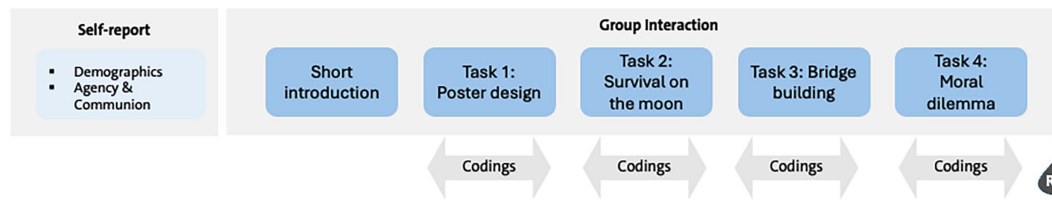
Participants first completed an online questionnaire assessing socio-demographic variables and personality constructs. Once completed, they were invited to the laboratory to take part in a group interaction that lasted around 2.5 hr. Participants were randomly assigned to same-gender groups. The meeting was structured as follows (see [Figure 2](#)): Participants first presented themselves briefly to the other group members. Next, the group completed a fixed sequence of four tasks. In the first task, the group created a poster detailing ideas on how to structure an ideal lecture (15 min). Task 2 featured the survival on the moon task (10 min), in which participants were asked to imagine that they are a space crew that was forced to land on the moon with the mission to reach the mother ship to ensure survival. Participants rank ordered 15 items in terms of their importance for the survival mission (Hall & Watson, 1970). In Task 3, participants received a box full of small plastic bricks and were instructed to build a bridge as long as possible in 15 min. Finally, Task 4 contained a moral dilemma wherein the group discussed whether it would be right to shoot down a passenger airline destined to be crashed into a football stadium (10 min). After completing all four tasks, participants rated their own experiences and, in a round-robin, how much they liked each of the other group members, how much they believed to be liked by them, as well as their own performance. All four tasks were recorded on video and used for behavior coding at person and group levels after data collection was completed.

Participants

In total, $N = 439$ students participated in 110 same-sex groups (56 all-female groups). Two groups had incomplete data (i.e., one person did not show up for the scheduled interaction, and one dropped out of the study), and we excluded these groups from the

⁴ The study design of the primary data collection was not preregistered.

Figure 2
Overview of Study Design



Note. All four tasks were recorded on video. Individual behaviors, group behaviors, and task performance were coded for each task separately after data collection was completed. RR = round-robin rating. See the online article for the color version of this figure.

analyses. The final sample consisted of $N = 432$ participants in 108 groups of four ($M_{\text{age}} = 26.61$, $SD = 5.97$, range = 18–62; 51% female). Most participants were enrolled in humanities (20.60%), followed by economics (12.73%), natural sciences (11.11%), medicine or health fields (10.19%), and social sciences (9.49%).

To our knowledge, there is no established method for conducting power simulations for GAPIMs. Various model assumptions must be considered, and little research exists to guide the specifics of these assumptions. Previous studies using GAPIMs with different predictors (e.g., gender; Garcia et al., 2015) show that relatively small samples ($N = 153$ in groups of five) yield meaningful results. Bonito (2022) suggested that to detect medium-sized actor similarity effects (0.30, with an intraclass correlation [ICC] of 0.10), groups should ideally consist of at least four participants, with a total of more than 75 groups. These findings suggest that our sample (108 groups of four people) has sufficient power to address our research questions.

Measures

Personality Characteristics

To assess agency and communion, we used the self-reported personality trait measure Big Five Inventory–2 (German version; Danner et al., 2016) measuring the Big Five traits on a scale of 1 (*strongly disagree*) to 5 (*strongly agree*). In the original scoring, the Big Five Inventory–2 items map onto the Big Five factors. Because agency and communion represent broader hierarchical levels of personality than the Big Five, they can be measured with an appropriate selection of items of the Big Five Inventory–2 (Digman, 1997; Entringer et al., 2022). In the current research, we adopted an item selection proposed by Entringer et al. (2022), who used the ant colony method to identify six items belonging to the domain of agency and seven items belonging to the domain of communion (see Supplemental Table OS2 for specific items). Entringer et al., 2022 found that the extracted dimensions of agency and communion strongly correlate with established measures of these constructs, with correlations comparable to those observed between the standard scales themselves. Moreover, this approach has been successfully validated through empirical data (Boileau et al., 2024; Gebauer et al., 2020). In the present data, internal consistencies were good for both scales ($\omega = .79$ for agency, $\omega = .80$ for communion).

Person-Level Outcomes

Individual Behaviors. Conflict, cooperation, communication, and coordination behaviors were assessed for each participant in each of the four group tasks by three external observers on a scale of

1 to 6. In total, six observers participated in a comprehensive training session that included an introduction to coding principles, potential biases in interpersonal evaluations, and extensive practice with pilot videos to ensure familiarity with the rating procedures. To standardize the process, observers also received a detailed coding manual. Observers were instructed to begin their observations in the first third of the recorded task. They were allowed to fast-forward or rewind the video. Each observer coded a fixed timeframe of 2–3 min.⁵ Observers' responses were assessed with standard questionnaire software.

Next, all six observers were assigned to a nested coding scheme: Three observers rated cooperation and communication behaviors for each participant in the first group task (poster design), while the other three observers rated coordination and conflict behaviors. After completing these ratings, the coding assignments were switched for the second task (moon landing). This sequence of coding assignments was then repeated for the third task (bridge building) and the fourth task (moral dilemma). Thus, each rater coded every participant four times but focused on different behaviors in each coding round. The interrater agreement was satisfactory for aggregates across the three raters within the separate tasks ($ICC[3, 3] = .40-.73$; see Supplemental Table OS3 for details), and the internal consistencies were high for aggregates across the four group tasks ($ICC[3, 4] = .63-.76$). Analyses with the individual observer-coded behaviors for each task were not included in our preregistration (see Supplemental Table OS1).

Individual Experiences. We assessed two types of perceptions. First, liking and metaliking (i.e., the belief of being liked by others) were assessed with the items “How much do you like this person?” and “How much do you think this person likes you?” on scales of 1–6 in the round-robin after all four group tasks were completed. That is, liking and metaliking perceptions represented directed dyadic ratings that each person completed about each interaction partner (Wong et al., 2024). We used these data in two ways. As preregistered, we conducted social relation model analyses (Kenny, 1994) to disentangle the three components that make up these interpersonal perceptions: a perceiver effect, a target effect, and a relationship effect. For our analyses at the person level, we used the perceiver effect of the actor's ratings (indicating how much that person liked their interaction partners in general) and the target

⁵ Because resources were limited, rating these shorter video snippets was more feasible instead of coding the full video for each participant in each task. This approach aligns with research showing that thin-slice assessments are valid indicators of behavior (Murphy & Hall, 2021).

effect of liking (indicating how much the actor was generally liked by others in the group) as outcome variables. Similarly, we considered the perceiver effect of metaliking—indicating how much actors thought they were liked by their interaction partners in general. Additionally, we considered liking and metaliking as dyadic outcomes in an exploratory analysis using the GAPIM-D (see [Supplemental Material](#) Section D).

Second, people were asked to rate their group's cohesion (e.g., "There is a strong sense of belonging between the team members") and potency (e.g., "My team can successfully solve a wide range of problems") on a scale of 1–6 after all four group tasks had been completed with three questions each ($\omega = .87-.90$).⁶ In addition to these preregistered indicators of a person's experiences, we aggregated all four indicators (liking, metaliking, perceived group cohesion, perceived group potency) into one composite score ($\omega = .88$).

Individual Performance. Participants answered two items assessing their own performance (i.e., "To what degree did you show a good performance?" and "To what degree were you effective?") on a scale of 1–6 after all four group tasks. Additionally, participants were asked to assess how much influence they had on the group and to what extent they provided valuable contributions on a scale of 1–6. The respective items were moderately to strongly correlated ($r = .42-.48$) and aggregated into a single score capturing a participant's performance or influence, respectively.

Group-Level Outcomes

Group Behaviors. In each of the four group tasks, observers coded the four behavior domains (conflict, cooperation, communication, coordination) on a scale of 1–6 at the group level. In total, 12 independent observers rated group behaviors. Coding procedures were parallel to person-level behavior coding: First, all observers participated in an extensive training session and a detailed coding manual. Next, all 12 observers followed a rotation scheme (i.e., the first six observers coded group cooperation and communication behavior in the first and third tasks, while the other six observers coded coordination and conflict behaviors in the first and third tasks; this order was switched for the second and fourth tasks).⁷

Similar to individual behaviors, we considered observer codings separately for each task as well as aggregated indicators across the four tasks per group. Interrater reliabilities at the level of the individual tasks were mostly satisfactory ($ICC[3, 6] = .49-.82$; [Supplemental Table OS3](#)),⁸ and internal consistencies of group behaviors across tasks were good ($ICC[3, 4] = .72-.80$).

Group Experiences. To assess the group's overall experience, we aggregated individuals' self-perceptions of group cohesion and group potency within each group. As an alternative indicator of the groups' experience, we additionally computed a group liking and group metaliking score (i.e., how much people in the group liked each other and thought to be liked on average). In parallel to the person level, we aggregated these four indicators (group cohesion, group potency, group liking, and group metaliking) to one overall indicator of a group's experience ($\omega = .91$).

Group Performance. In each of the four tasks, group performance was assessed either through observer codings or objective measurements. Performance in Task 1 (poster design) was rated by three observers judging the poster design and content ($ICC[3, 3] = .97$). In Task 2 (survival on the moon), we used an objective

performance criterion to assess how closely the group's decisions aligned with the optimal solution. In Task 3 (bridge building), performance was also objectively measured by assessing the length of the bridge in cm. Finally, in Task 4 (moral dilemma), three observers rated the arguments of the group ($ICC[3, 3] = .94$).

Control Variables

Sociodemographic information was assessed in the questionnaire before the group interaction. Participants reported their age and gender, which was coded 0 for women and 1 for men.

Determining GAPIM Predictors at Person and Group Levels

We conducted several preprocessing steps ([Garcia et al., 2015](#)) to derive the composition variables for the GAPIM-I and GAPIM-G. The formulas used in each step are described in [Table 1](#). In the first step, we effect coded agency and communion scores for each person. In the second step, we derived three variables for each trait domain based on the effect-coded variables in Step 1, which represent the core elements of the GAPIM-I to predict outcomes at the person level. First, the group(−1) personality is the mean of the effect-coded agency (or communion) of all group members excluding person i . Second, the actor-group fit represents the product term between the actor and group(−1) personality variables. Third, the group(−1) similarity captures the similarity among all interaction partners in the group excluding person i , reflected in the formula in [Table 1](#) (see [Kenny & Garcia, 2012](#), for details).

In summary, every GAPIM variable at the person level is based on effect-coded agency and communion scores that fall within the range of −1 to +1. This structure was initially designed to analyze dichotomous composition variables, such as gender ([Kenny & Garcia, 2012](#)), and simplifies the interpretation of the product terms. That is, participants from one category would receive a score of +1 (e.g., female), while those from the other category would receive a score of −1 (e.g., male). When dealing with continuous variables like personality scores, the effect coding procedure is modified. Here, the highest (agency or communion) score in the sample is recoded as +1, while the lowest score is recoded as −1. Variables computed at the dyadic level for GAPIM-D analyses are detailed in the [Supplemental Material](#).

Analyses

The model specifications of the GAPIM depend on the level of the outcome variable (person vs. group). At each level, we fitted two models per outcome: one with agency and one with communion composition variables. In the following, we first describe the setup of the GAPIM-I and then the GAPIM-G. We report exact p values and discuss all effects significant at $p < .05$. Additionally, we provide adjusted p values with a false discovery rate to account for

⁶ See our preregistration for all items.

⁷ Observers either coded person- or group-level behaviors and never both levels.

⁸ One exception was a low interrater reliability for conflict behavior in Task 1 (poster design; $ICC[3, 6] = .16$); we are therefore cautious in interpreting all group-level results for conflict behavior in this task.

Table 1*Computation of the GAPIM Predictors*

Variable	Formula	Explanation of terms	Interpretation
GAPIM-I for person-level outcomes			
1. Actor personality	$X_{ik} = \frac{2(X_{rawik} - X_{raws})}{(X_{rawL} - X_{raws})} - 1$	X_{ik}: effect-coded score of person i in group k X_{rawik}: the raw personality score of person i in group k (e.g., their self-perceived level of agency or communion) X_{raws} and X_{rawL}: the smallest and largest value in the sample, respectively	Positive values (up to +1) indicate that the actor scores high on agency or communion; negative values (up to -1) indicate that the actor scores low on agency or communion compared with the values in the entire sample.
2. Group(-1) personality	$X'_{ik} = \frac{n_k \times \bar{X}_k - X_{ik}}{n_k - 1}$	X'_{ik}: partners' personality for person i in group k (i.e., average level of agency or communion excluding the actor) n_k: group size of group k $\bar{X}_k$: averaged actor personality in group k	Positive values (up to +1) indicate that the actor's fellow group members on average score high on agency or communion; negative values (up to -1) indicate that the actor's fellow group members have a low average personality level.
3. Actor-group fit	$I_{ik} = X_{ik} \times X'_{ik}$	I_{ik}: actor-group fit of person i in group k (i.e., product term of actor and partners' personality)	Positive values (up to +1) indicate that the actor and the rest of the group have similar agency or communion levels; negative values (up to -1) indicate a deviation between the actor's own personality and the average personality of the actor's fellow group members.
4. Group(-1) similarity	$I'_{ik} = \frac{n_k \times \bar{X}_k^2 - \sigma^2(X_k) - 2 \times I_{ik}}{n_k - 2}$	I'_{ik}: partners' similarity for person i in group k, interaction of all pairs of the others' effects or how similar the other $n - 1$ members of group k are to each other $n - 2$ members	Positive values (up to +1) reflect more similarity, and more negative values (up to -1) indicate more diversity among an actor's fellow group members.
GAPIM-G for group-level outcomes			
5. Group personality	$\bar{X}_k$	$\bar{X}_k$: average personality level of the n members of group k	Positive values (up to +1) indicate that the group on average scores high on agency or communion; negative values (up to -1) indicate that the group scores low on agency or communion.
6. Group similarity	$\bar{I}_k$	$\bar{I}_k$: the average similarity of all $n(n - 1)/2$ pairs of members of group k, that is, the group mean of partners' similarity I'_{ik}	Positive values (up to +1) reflect more similarity, and more negative values (up to -1) indicate more diversity among all group members.

Note. To ease interpretation, we revised the labeling of effects from [Kenny and Garcia \(2012\)](#). In their work, they labeled the Group(-1) personality as the others' effect, actor-group fit as actor similarity, and group similarity as the others' similarity. GAPIM = group actor-partner interdependence model.

the multiple tests performed and discuss effects that do not withstand this correction procedure ([Benjamini & Hochberg, 1995](#)).

GAPIM-I for Person-Level Outcomes

The GAPIM-I is a multilevel model with two levels reflecting persons nested within groups. The model includes five fixed-effects parameters: the intercept, the actor personality, the partners' personality, actor-group fit, and partner similarity, detailed in the data processing section. For example, in the model focusing on agency composition predicting a person's conflict behavior, the within-person level of the GAPIM-I is expressed as follows:

Level 1:

$$\begin{aligned}
 \text{conflict}_{ik} = & b_{0k} + b_1(\text{actor personality})_{ik} \\
 & + b_2(\text{partners' personality})_{ik} \\
 & + b_3(\text{actor-group fit})_{ik} \\
 & + b_4(\text{partner similarity})_{ik} + e_{ik}, \quad (1)
 \end{aligned}$$

Level 2:

$$b_{0k} = \gamma_{00} + u_{0k}. \quad (2)$$

In this model, the term b_{0k} describes the extent to which different groups exhibit different levels of conflict behavior. Specifically, the term represents the group-level intercepts, which are modeled as random effects that are assumed to be normally distributed with a mean of 0 and variance τ_{00} . The other b parameters can be interpreted as follows: b_1 represents the effect of the actor's personality, indicating how person i 's level of agency is linked to how much they engage in conflict behavior during the interaction. The parameter b_2 reflects the effect of partners' personality, describing the link between the average level of agency of the other group members (excluding person i) and how conflictual person i behaves. The parameter b_3 captures the actor-group fit, that is, the link between the similarity among person i and other group members' agency and person i 's conflict behavior. Reflecting the final GAPIM-I predictor, the parameter b_4 reflects the effect of partner similarity, indicating how the average similarity of agency among interaction partners (excluding person i) relates to

person i 's conflict behavior. Last, the term e_{ik} captures the residual at the individual level, reflecting the deviation from conflict behavior predicted by the intercept and the predictors in the model (it is assumed to follow a normal distribution with mean 0 and variance σ^2). As indicated by the equation at Level 2, the GAPIM-I does not include random slopes of the GAPIM terms.

We conducted several exploratory extensions of the original GAPIM-I, which were not preregistered (see [Supplemental Table OS1](#)). First, we expanded all models to include a person's age at Level 1 and the group's gender at Level 2 as control variables. We decided to incorporate these control variables across all models based on preanalytical results illustrating several substantial correlations with outcome variables ([Supplemental Table OS4](#)).⁹ Furthermore, we considered that there might be an additional interaction effect between partners' personalities and partner similarity. For example, high partner similarity in terms of agency might affect an actor's communication behavior only if this similarity is observed on a generally high level (all partners high on agency) but less on a generally low level (all partners low on agency). Finally, we conducted an additional robustness check by fitting combined models with both agency and communion composition components.

GAPIM-G for Group-Level Outcomes

To investigate composition effects of agency and communion on group-level outcomes, we used the following model:

$$\text{conflict}_k = b_0 + b_1(\text{group personality})_{ik} + b_2(\text{group similarity})_k + e_k. \quad (3)$$

Returning to the example of agency and conflict behavior, b_1 reflects the link between the average level of agency within group k and the observed level of conflict within the group. The term b_2 describes the extent to which the similarity of agency within the group predicts the amount of conflict there is in the group. We expanded these models in parallel to the GAPIM-Is: Full models included the group-level control variables group mean age and gender. Additionally, we explored a possible interaction between group personality and group similarity to examine whether groups with high average levels of agency (or communion) and a high degree of similarity would behave or perform differently compared with groups with high average agency (or communion) but lower levels of similarity. Analogous to the person level, we ran combined models (including both agency and communion composition) as a robustness check.

Results

We begin by discussing the results of the GAPIM-Is, which explore how agency and communion composition among group members predict individual behaviors, experiences, and performances. Next, we present the findings from the GAPIM-Gs, which illustrate how the composition of agency and communion shapes outcomes at the group level. In presenting the results, we focus on general effect patterns across tasks. However, because some results differed substantially across the four tasks, we included results at the task level in the main tables and highlight noteworthy effects for specific tasks in the text. Results of the exploratory GAPIM-D analyses are reported in the [Supplemental Material](#).¹⁰

Composition Effects on Person-Level Outcomes

[Tables 2](#) and [3](#) summarize the main results of the GAPIM-Is in the domains of agency and communion, respectively. The [Supplemental Material](#) contains additional detailed information, such as correlations of all person-level variables (i.e., GAPIM-I predictors and all outcome variables) in [Supplemental Tables OS4 and OS5](#).¹¹ Additionally, [Supplemental Table OS7](#) shows covariate effects of age and gender, [Supplemental Table OS8](#) summarizes the explained variance (R^2) for all models, and [Supplemental Tables OS9 and OS10](#) include results of models without covariates (as preregistered).

Across all GAPIM-Is, three global aspects stood out. First, the ICC coefficients from the models indicated that most of the variance in outcomes could be attributed to differences among people within the same groups rather than to systematic differences between groups. Second, result patterns varied significantly across the four tasks in the interaction (upper part of [Tables 2](#) and [3](#)). Third, robustness checks (i.e., models with and without covariates, models with combined agency and communion effects instead of separate GAPIM-Is; [Supplemental Tables OS8–OS10](#)) yielded highly similar results to the models reported in the main text.

Actor Personality

As shown on the left-hand side in [Table 2](#), the actor's personality—people's own levels of agency and communion—emerged as the strongest and most consistent predictor of their behaviors and self-rated performance. High agency showed largely positive effects: Highly agentic people communicated more and coordinated better with others across tasks ($bs = 0.37, ps < .001$). They also perceived their performance and influence on the group to be greater than those scoring lower in agency ($bs = 0.24–0.33, ps < .002$).

As illustrated on the left-hand side in [Table 3](#), actor communion effects were beneficial mainly for people's conflict behavior and individual experiences. That is, people with higher communion showed less conflict behavior across tasks ($b = -0.14, p < .001$) and generally reported a more positive experience ($b = 0.42, p < .001$). Additionally, highly communal individuals believed they performed better ($b = 0.22, p = .003$).

Group(–1) Personality

The most notable group(–1) effects emerged for agency on conflict behavior: When interaction partners were highly agentic, actors were

⁹ For transparency, the full model equations for the GAPIM-I and GAPIM-G including covariates are shown in [Supplemental Table OS6](#). Results of the preregistered models without covariates were highly similar and are reported in [Supplemental Tables OS8 and OS9](#) displaying the GAPIM-I at the individual level and [Supplemental Table OS16](#) at the group level.

¹⁰ The GAPIM-D analyses focus on liking and metaliking as the only dyadic data collected in the round-robin.

¹¹ Given the relatively high correlations among composition components at the person level (see [Supplemental Table OS5](#)), we performed an additional robustness check where we centered the GAPIM-I variables (using typical sample-mean centering) to compare findings with the effect coding. The results of the models including centered variables were very similar to the preregistered models with effect-coded variables. The code for the robustness analyses is available on the Open Science Framework at <https://osf.io/muaq5/>.

Table 2
Agency Composition Effects Predicting Person-Level Outcomes

Model	Actor personality			Group(-1) personality			Actor-group fit			Group(-1) similarity		
	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>
Individual behaviors												
Conflict	0.03	0.04	.511	-0.20	0.07	.006	0.46	0.18	.010	0.25	0.18	.165
Task 1: Poster	0.00	0.04	.910	-0.13	0.06	.041 [†]	0.03	0.17	.840	0.23	0.17	.186
Task 2: SoM	0.07	0.06	.235	-0.02	0.11	.844	0.79	0.27	.003	-0.11	0.27	.676
Task 3: Bridge	-0.01	0.03	.785	-0.24	0.05	<.001	0.29	0.14	.045 [†]	0.29	0.14	.044 [†]
Task 4: MD	0.04	0.09	.675	-0.40	0.16	.015 [†]	0.74	0.40	.065	0.59	0.40	.142
Cooperation	0.14	0.08	.065	0.09	0.13	.463	0.23	0.34	.500	-0.52	0.34	.126
Task 1: Poster	0.34	0.12	.004	0.05	0.20	.783	0.34	0.53	.513	-1.28	0.53	.015
Task 2: SoM	-0.03	0.10	.787	-0.07	0.17	.684	1.13	0.45	.013	-1.06	0.45	.019
Task 3: Bridge	0.10	0.09	.258	0.14	0.15	.361	0.01	0.40	.986	0.22	0.40	.576
Task 4: MD	0.15	0.12	.205	0.25	0.19	.187	-0.58	0.51	.257	0.05	0.51	.916
Communication	0.37	0.10	<.001	-0.06	0.16	.729	0.30	0.43	.479	-0.23	0.43	.585
Task 1: Poster	0.41	0.12	.001	-0.01	0.20	.966	0.42	0.55	.450	-0.74	0.55	.183
Task 2: SoM	0.10	0.10	.320	-0.04	0.17	.815	0.95	0.46	.038 [†]	-0.88	0.46	.055
Task 3: Bridge	0.57	0.15	.000	-0.18	0.25	.471	0.51	0.66	.442	0.29	0.66	.658
Task 4: MD	0.38	0.13	.005	0.00	0.22	.993	-0.64	0.59	.277	0.34	0.59	.564
Coordination	0.37	0.09	<.001	-0.04	0.16	.812	0.00	0.42	.998	-0.23	0.42	.578
Task 1: Poster	0.51	0.12	<.001	0.11	0.19	.578	0.18	0.52	.722	-0.91	0.52	.080
Task 2: SoM	0.42	0.14	.002	-0.30	0.22	.182	0.27	0.60	.653	-0.20	0.60	.738
Task 3: Bridge	0.31	0.10	.001	0.06	0.16	.720	-0.17	0.43	.687	0.13	0.43	.756
Task 4: MD	0.24	0.15	.115	-0.02	0.25	.941	-0.24	0.66	.714	0.06	0.66	.933
Individual experiences												
Overall experience	0.17	0.09	.080	0.28	0.18	.126	0.86	0.44	.052	-0.44	0.44	.316
Liking others	0.03	0.12	.798	0.27	0.21	.208	1.13	0.54	.037 [†]	-0.72	0.54	.183
Being liked	-0.01	0.11	.910	0.29	0.21	.157	0.93	0.51	.071	-0.50	0.51	.335
Metaliking	0.27	0.11	.013 [†]	0.25	0.18	.156	0.27	0.47	.566	-0.32	0.47	.501
Cohesion	0.32	0.14	.016 [†]	0.45	0.25	.068	0.95	0.62	.119	-0.56	0.62	.358
Potency	0.22	0.14	.102	0.14	0.26	.596	1.03	0.65	.110	-0.11	0.65	.867
Individual performance												
Performance	0.24	0.08	.002	0.03	0.14	.855	0.19	0.35	.578	0.18	0.35	.600
Influence	0.33	0.07	<.001	-0.06	0.12	.600	-0.06	0.32	.846	0.40	0.32	.215

Note. Values presented in bold indicate statistical significance at $p < .05$. To provide an overview at first glance, we color coded the effects: Dark green highlights robust positive effects, while light green signifies positive effects that did not reach significance after adjusting for multiple testing. By contrast, dark red indicates significant negative effects, with light red representing negative effects that were not significant after adjustment. SoM = survival on the moon; MD = moral dilemma; SE = standard error. See the online article for the color version of this table.

[†] $p < .05$ indicates that effects were significant but did not remain significant after adjusting for multiple testing.

less likely to engage in conflicts ($b = -0.21, p = .005$). For communion, only one group(-1) effect was robustly significant (i.e., survived correction for multiple testing) at the task level: People who had partners with higher average communion levels coordinated better ($b = 0.56, p = .017$) when discussing the survival on the moon mission (Task 2).

Actor-Group Fit

Across tasks, only one robustly significant actor-group fit effect (i.e., the interaction between actor and partners' personality trait levels) emerged for agency and individual conflict behavior ($b = 0.47, p = .010$; see the column "Actor-group fit" in Table 2). Figure 3 illustrates this effect: People showed the most conflict behavior when their own agency level was similar to that of the other group members, particularly when both the actor and the group scored high on agency.

While no actor-group interactions were found for individual experiences of liking and metaliking, the additional exploratory GAPIM-D analysis at the dyadic level (see Supplemental Tables OS23 and OS24 in Section D) uncovered a significant actor-partner fit in terms of agency: When actors and specific interaction partners reported similar

levels of agency, actors tended to report liking that partner more and believed that they were liked more by that partner.

Group(-1) Similarity

No similarity effects emerged consistently across tasks for either agency or communion. A more nuanced picture emerged in our additional analysis. Looking at individual tasks, as shown in the upper part of Table 2, more agency diversity among interaction partners seemed to be beneficial for people's cooperative behavior when they designed a poster (Task 1, $b = -1.28, p = .015$) and discussed the survival mission on the moon (Task 2, $b = -1.06, p = .019$). Moreover, the exploratory models featuring an additional product term between the group(-1) personality and group(-1) similarity (Supplemental Table OS9 and OS10) showed several negative interaction effects on actors' behaviors: If interaction partners scored homogeneously low on communion, actors behaved less cooperative, communicative, and coordinated across tasks ($b_{\text{low_com}} = -2.58$ to -2.98), while there were few behavioral differences when interaction partners scored homogeneously high

Table 3*Communion Composition Effects Predicting Person-Level Outcomes*

Model	Actor personality			Group(−1) personality			Actor-group fit			Group(−1) similarity		
	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>
Individual behaviors												
Conflict	−0.14	0.04	<.001	−0.07	0.08	.383	0.1	0.12	.375	−0.08	0.12	.515
Task 1: Poster	−0.08	0.04	.046 [†]	−0.02	0.07	.760	0.05	0.11	.681	−0.11	0.11	.327
Task 2: SoM	−0.15	0.06	.010	−0.09	0.11	.442	0.29	0.18	.101	−0.01	0.18	.977
Task 3: Bridge	−0.06	0.03	.071	−0.06	0.06	.291	0.1	0.1	.312	−0.03	0.1	.793
Task 4: MD	−0.27	0.08	.002	−0.10	0.17	.579	−0.02	0.26	.953	−0.17	0.26	.520
Cooperation	0.07	0.07	.380	0.28	0.13	.037[†]	0.25	0.22	.249	−0.44	0.22	.047[†]
Task 1: Poster	0.07	0.12	.612	0.32	0.21	.120	0.31	0.35	.375	−0.34	0.35	.324
Task 2: SoM	0.13	0.10	.205	0.29	0.18	.098	0.03	0.3	.909	−0.56	0.3	.060
Task 3: Bridge	0.12	0.09	.169	0.28	0.16	.083	−0.08	0.26	.764	−0.12	0.26	.637
Task 4: MD	−0.04	0.11	.668	0.21	0.20	.302	0.74	0.33	.026[†]	−0.71	0.33	.033[†]
Communication	−0.11	0.09	.254	0.29	0.17	.084	0.29	0.28	.306	−0.37	0.28	.191
Task 1: Poster	0.00	0.12	.941	0.24	0.22	.264	0.44	0.36	.229	−0.39	0.36	.280
Task 2: SoM	−0.19	0.10	.061	0.18	0.18	.316	0.49	0.3	.104	−0.18	0.3	.545
Task 3: Bridge	−0.07	0.15	.626	0.51	0.26	.053	−0.23	0.44	.593	−0.36	0.44	.410
Task 4: MD	−0.17	0.13	.195	0.24	0.23	.290	0.47	0.39	.221	−0.56	0.39	.146
Coordination	−0.10	0.09	.258	0.29	0.16	.083	0.37	0.28	.180	−0.63	0.28	.023
Task 1: Poster	−0.16	0.12	.173	0.07	0.21	.723	0.56	0.35	.110	−0.23	0.35	.517
Task 2: SoM	−0.12	0.13	.363	0.56	0.23	.017	0.68	0.39	.084	−1.37	0.39	.001
Task 3: Bridge	0.06	0.09	.560	0.37	0.17	.026[†]	−0.14	0.28	.608	−0.24	0.28	.392
Task 4: MD	−0.19	0.14	.189	0.14	0.26	.596	0.38	0.43	.380	−0.69	0.43	.110
Individual experiences												
Overall experience	0.42	0.09	<.001	0.02	0.19	.903	−0.25	0.29	.384	0.38	0.29	.189
Liking others	0.47	0.12	<.001	0.1	0.22	.650	−0.48	0.35	.166	0.42	0.35	.228
Being liked	0.23	0.11	.041 [†]	0.34	0.21	.113	−0.24	0.33	.474	0.20	0.34	.548
Metaliking	0.46	0.10	<.001	−0.08	0.18	.653	−0.12	0.3	.698	0.29	0.30	.345
Cohesion	0.36	0.13	.006	−0.23	0.26	.368	0.43	0.4	.285	0.56	0.40	.163
Potency	0.56	0.14	<.001	−0.01	0.27	.972	−0.86	0.42	.041[†]	0.46	0.42	.273
Individual performance												
Performance	0.22	0.08	.003	−0.01	0.14	.941	−0.22	0.23	.332	0.05	0.23	.842
Influence	0.06	0.07	.425	0.07	0.12	.558	−0.20	0.22	.365	−0.04	0.22	.872

Note. Values presented in bold indicate statistical significance at $p < .05$. To provide an overview at first glance, we color coded the effects: Dark green highlights robust positive effects, while light green signifies positive effects that did not reach significance after adjusting for multiple testing. By contrast, dark red indicates significant negative effects, with light red representing negative effects that were not significant after adjustment. SoM = survival on the moon; MD = moral dilemma; SE = standard error. See the online article for the color version of this table.

[†] $p < .05$ indicates that effects were significant but did not remain significant after adjusting for multiple testing.

on communion ($b_{\text{high_com}} = 0.41\text{--}0.66$). Figure OS1 illustrates these interaction effects between group communion level and similarity.

Covariates: Age and Gender Effects

Across all GAPIM-Is, we observed consistent effects of age and gender (Supplemental Table OS6). In general, older people tended to be slightly more conflictual and less cooperative, communicative, and coordinated. Additionally, older people tended to be liked less by other group members. While gender showed no relation to individual conflict behavior, female participants generally displayed greater cooperation, communication, and coordination in their interactions compared with male participants. Furthermore, the level of liking was greater in female groups compared with male groups.

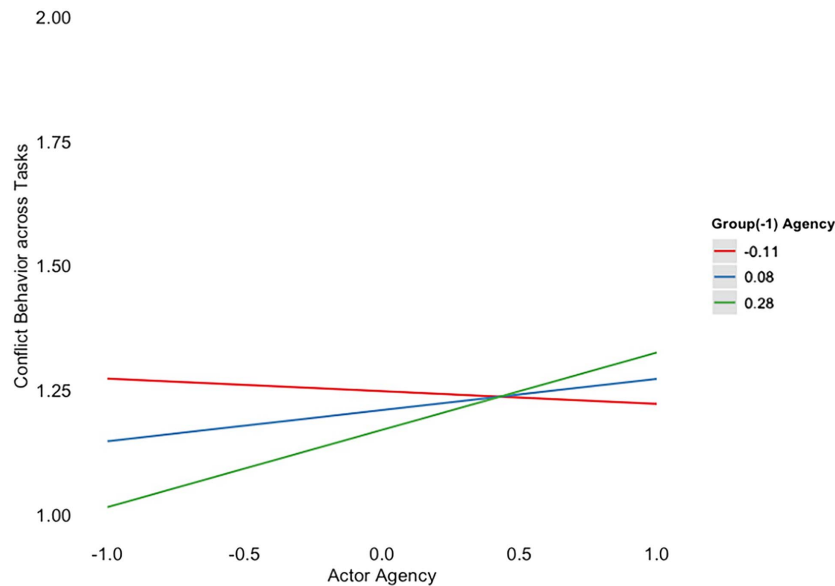
Composition Effects on Group-Level Outcomes

In the group-level models (GAPIM-G), we examined two personality composition components: the group's average personality and group similarity. Supplemental Tables OS14 and OS15 present

the correlations with all group-level outcome variables. Focusing on the group-level GAPIM predictors, groups with higher agency or communion averages were likely to be more similar in terms of the two traits.¹² To validate the group similarity measure, we correlated it with the within-group standard deviation for each trait. As expected, increased group similarity correlated with less variation in agency or communion, as indicated by lower standard deviations (agency: $r = -.37$; communion: $r = -.52$). At the same time, the nonperfect correlations suggest that considering all dyadic similarities in a group adds nuance above a global index of dispersion among individuals. Table 4 summarizes the main findings of the GAPIM-Gs. The effects of group age and gender as covariates, along with R^2 values for model fit and model results without covariates are detailed in Supplemental Tables OS16–OS18.

¹² This positive correlation may occur because groups scoring at the higher (or lower) end of the rating scale naturally show less variability (i.e., when measurements approach the upper limits of a rating scale, there is mathematically less room for variability among responses).

Figure 3
Agency Actor-Group Fit on Individual Conflict Behavior Across Tasks



Note. The colored lines depict the effect of actor agency on conflict behavior at the average level of group(-1) agency (blue line), at one standard deviation below the mean (red line), and at one standard deviation above the mean (green line). See the online article for the color version of this figure.

Group Personality

There were four findings regarding the groups' overall personality level on group outcomes that stood out. First, across tasks, higher levels of agency within a group were associated with less group conflict ($b = -0.33, p = .027$) and more cooperation ($b = 0.48, p = .022$). Looking at the separate tasks, both effects only appeared to be substantial in the fourth task, where groups solved a moral dilemma together. Second, in terms of the group's experiences, groups with higher overall agency felt more cohesive ($b = 0.73, p = .013$) and believed they were liked by their peers ($b = 0.50, p = .013$). Third, the group's communion level was linked to their conflict behavior, but not to their cooperation, communication, or coordination behavior ($b = -0.40, p = .012$). However, even the effect for conflict behavior was not particularly robust, as it did not survive adjustment for multiple testing and emerged only when group communion level and similarity were both predictors in the model, but not as a zero-order effect. Fourth, as shown in Table 4, neither the group's agency nor communion levels linked to performance in any of the four tasks.

Group Similarity

Beyond the effect of the group's average personality, group similarity effects only emerged in the domain of group conflict behavior. Focusing on agency, groups with similar agency levels were more likely to engage in conflict ($b = 1.28, p = .003$), particularly during the survival on the moon task and while discussing the moral dilemma. Our exploratory analysis including an additional interaction term between the group's average level and group similarity (Supplemental Table OS16) showed that conflict

peaked when there was a low overall level of agency in the group and high similarity at the same time, thus when all group members scored low on agency.

Focusing on communion, the main models indicated that beyond the negative effect of a group's average communion, groups that were more similar were perceived as more conflictual in Task 1, where they designed a poster ($b = 0.48, p = .029$), and in Task 2, where they discussed the survival on the moon mission ($b = 1.33, p = .018$). Notably, because the interrater agreement on conflict behavior in Task 1 was quite low, we caution against overinterpreting this effect. The exploratory models for communion (Supplemental Table OS17) revealed one significant interaction effect on metaliking: When all group members scored high on communion (high average and high similarity), the group generally believed that people liked each other, while groups wherein all members scored low on communion (low average and high similarity) were more pessimistic about being liked.

Covariates: Age and Gender Effects

We observed small but consistent effects of age and gender at the group level, similar to what we found at the person level. Groups that were older on average tended to have slightly more conflict and were less cooperative and coordinated. In terms of performance, older groups generally performed worse in the poster design task (Task 1) and while discussing the moral dilemma (Task 4), according to observer assessments. In terms of gender, all-female groups typically demonstrated greater cooperation, communication, coordination, and overall cohesion. Although there was some indication that all-male groups were more conflictual, these effects were not consistent across models and tasks. Notably, the only significant gender difference related to

Table 4*Agency and Communion Composition Effects on Group-Level Outcomes*

Model	Agency						Communion					
	Group average			Group similarity			Group average			Group similarity		
	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>	<i>b</i>	<i>SE</i>	<i>p</i>
Group behaviors												
Conflict	−0.33	0.15	.027	1.28	0.42	.003	−0.41	0.16	.013	0.32	0.30	.282
Task 1: Poster	−0.20	0.11	.076	−0.19	0.32	.544	−0.46	0.11	<.001	0.46	0.21	.029
Task 2: SoM	−0.28	0.26	.283	2.64	0.75	.001	−0.76	0.29	.011	1.29	0.53	.018
Task 3: Bridge	−0.28	0.16	.078	0.84	0.45	.064	−0.24	0.17	.156	0.06	0.31	.855
Task 4: MD	−0.56	0.23	.017	1.85	0.66	.006	−0.18	0.26	.493	−0.53	0.47	.267
Cooperation	0.48	0.20	.022	−0.78	0.59	.184	0.35	0.22	.123	−0.10	0.41	.802
Task 1: Poster	0.33	0.27	.223	−0.59	0.77	.447	0.45	0.29	.125	−0.51	0.54	.348
Task 2: SoM	0.31	0.26	.243	−0.41	0.75	.588	0.54	0.28	.058	−0.71	0.52	.173
Task 3: Bridge	0.38	0.30	.205	−0.75	0.85	.376	0.20	0.32	.537	0.16	0.59	.784
Task 4: MD	0.88	0.29	.003	−1.38	0.84	.104	0.20	0.33	.535	0.64	0.60	.287
Communication	0.51	0.26	.051	0.28	0.73	.704	0.19	0.29	.513	−0.16	0.52	.756
Task 1: Poster	0.72	0.36	.050	0.21	1.04	.837	0.34	0.40	.402	−0.37	0.74	.615
Task 2: SoM	0.07	0.25	.774	0.82	0.72	.258	0.25	0.27	.370	−0.26	0.50	.607
Task 3: Bridge	0.48	0.38	.210	−0.30	1.10	.787	0.05	0.42	.907	0.24	0.77	.758
Task 4: MD	0.75	0.28	.010	0.39	0.82	.637	0.12	0.32	.720	−0.26	0.59	.664
Coordination	0.44	0.20	.031 [†]	−0.36	0.58	.536	0.20	0.22	.368	−0.13	0.41	.752
Task 1: Poster	0.42	0.30	.166	0.38	0.86	.661	0.15	0.33	.640	−0.26	0.60	.670
Task 2: SoM	0.32	0.27	.236	−0.63	0.76	.409	0.44	0.29	.134	−0.48	0.53	.365
Task 3: Bridge	0.44	0.27	.110	−0.27	0.78	.729	−0.08	0.30	.789	0.42	0.55	.451
Task 4: MD	0.60	0.28	.035 [†]	−0.91	0.80	.257	0.30	0.31	.336	−0.20	0.57	.730
Group experience												
Overall experience	0.46	0.22	<.001	0.46	0.64	.473	0.40	0.24	.100	0.23	0.44	.606
Group cohesion	0.73	0.29	.013	0.44	0.71	.536	0.11	0.31	.725	1.07	0.57	.064
Group potency	0.33	0.30	.286	0.45	0.82	.583	0.54	0.33	.106	−0.33	0.61	.592
Liking others	0.28	0.25	.261	0.97	0.87	.268	0.56	0.26	.032[†]	−0.03	0.48	.946
Metaliking	0.50	0.20	.013	−0.03	0.57	.963	0.37	0.22	.085	0.20	0.40	.617
Group performance												
Task 1: Poster	1.50	1.54	.330	5.07	4.4	.251	0.94	1.69	.580	−0.60	3.11	.849
Task 2: SoM	0.13	4.72	.978	1.53	13.52	.910	4.30	5.11	.402	−4.07	9.38	.666
Task 3: Bridge	−171.26	182.06	.349	141.12	522.62	.788	9.40	195.82	.962	204.96	360.56	.571
Task 4: MD	1.70	1.09	.122	0.00	3.13	.999	−1.95	1.18	.103	3.63	2.17	.098

Note. Values presented in bold indicate statistical significance at $p < .05$. To provide an overview at first glance, we color coded the effects: Dark green highlights robust positive effects, while light green signifies positive effects that did not reach significance after adjusting for multiple testing. By contrast, dark red indicates significant negative effects, with light red representing negative effects that were not significant after adjustment. SoM = survival on the moon; MD = moral dilemma; SE = standard error. See the online article for the color version of this table.

[†] $p < .05$ indicates that effects were significant but did not remain significant after adjusting for multiple testing.

performance was that male groups built longer bridges than female groups in Task 3.

Discussion

The present study investigated the role of personality composition effects in initial group interactions. Specifically, we used the GAPIM (Kenny & Garcia, 2012) as a framework to understand how the composition of two socially relevant personality traits—agency and communion—affects people and groups across four different tasks. Integrating the findings across levels, outcomes, and tasks, three broad insights emerged. First, people's self-perceived personality was a key predictor of their behaviors, experiences, and performances. That is, individuals with high agency took the lead by being more communicative and coordinated, while those high in communion viewed others in their group more positively. Second, these person-level effects did not simply translate to the group level. Here, higher average agency in groups promoted overall cooperation and cohesion but also linked to

more overall conflict, whereas high group communion was not robustly linked to any outcome domains. Overall, a group's average communion seemed to play a less important role compared to the group's level of agency. Third, we observed few composition effects across person and group levels. Most notably, people who were similar to their group in terms of agency were more engaged in conflicts, and groups whose members had similar agency levels were also more likely to experience conflicts as a whole. By contrast, neither agency or communion actor-group fit nor similarity measures related consistently to any other behavior, experience, or performance indicator across tasks.

Our research contributes to the understanding of personality composition in groups by unifying prior research in three regards. First, our results provide insights into the composition patterns of two broad personality traits, agency and communion, which can be compared and linked with earlier research on personality composition. Second, we provide a comprehensive framework to investigate multiple personality composition components at both person and group levels. Third, the

present findings provide, to our knowledge, the most comprehensive documentation of composition effects across a range of outcomes and across different tasks using a multimethod approach based on a large sample of small groups in the early stages of the acquaintance process.

The Role of Agency and Communion for Persons and Groups

Our findings indicate that people's behaviors, experiences, and self-perceived performance are primarily driven by their own personality traits, which resonates with previous work on dyadic interactions (Breil et al., 2019; Cuperman & Ickes, 2009) and largely aligns with studies investigating group settings (Gonzalez-Mulé et al., 2014; Prewett et al., 2018; Schmidt et al., 2012). We observed different roles of high agency and communion for person outcomes: Agency mostly contributed to task-related behaviors and performance perceptions, with individuals high in agency coordinating better and believing they performed well. On the contrary, communion predicted relationship-oriented perceptions, with individuals high in communion liking others more and viewing their group as more cohesive. Both findings are consistent with theoretical assumptions about the social functions of agency and communion, which are related to either getting ahead of others or getting along with others (Hopwood et al., 2021; Wiggins, 1979). In this regard, our findings add to the personality and organizational literature showing that people's stable agentic tendencies drive them to engage in behaviors that make them more likely to emerge as leaders in group settings (Grosz et al., 2020; Härtel et al., 2025). By contrast, those with strong communal tendencies seem to focus on promoting their own perceptions of group harmony, which may lead them to adopt supporting or listening roles in a group rather than actively taking the lead (cf. Badura et al., 2018).

Importantly, the effects of agency and communion observed at the person level were not reflected at the group level. That is, groups with higher average agency showed less conflict, more cooperation, and better cohesion at the group level, but they were not more communicative or coordinated (different to the person level). In addition, there was a tendency for groups with high average communion to exert less conflict (similar to the person level), but average group communion was otherwise unrelated to group outcomes. These discrepancies highlight that individual behaviors and experiences do not simply aggregate into group-level results (Moreland, 2013). This becomes particularly interesting when examining affective experiences at different levels: While a person's agency does not systematically link to liking others in the group, the group's overall agency seems to be important for the group's cohesion, which has also been suggested by prior research (van Vianen & De Dreu, 2001). In sum, our study is one of the first to comprehensively integrate person and group personality predictors and outcomes to understand the role of personality in social interactions at these different levels.

Evidence for Agency and Communion Composition Effects

Beyond exploring how a person's own personality and group average levels of agency and communion affect behaviors, experiences, and performance, one central aim of our research was to better understand if, how, and where the more complex composition effects of personality—that is, actor-group fit and group similarity—would manifest in group interactions. Across models, composition effects

consistently emerged in the domain of conflict behavior: At the person level, people experienced fewer conflicts when their agency level differed from the average agency of the group. At the group level, groups with a diversity of agency (i.e., low agency similarity) were perceived as less conflictual by external observers. In fact, our exploratory analyses indicated that groups in which everyone had low agency levels experienced the most conflict, suggesting that it is not the amount of diversity per se but rather the interplay of different levels of agency among group members that shapes conflict dynamics. Broadly, these findings support the idea of complementary agency proposed by person–environment fit perspectives (Kristof-Brown & Guay, 2011) and suggest that people and groups as a whole function better—in terms of fewer conflicts—when there is a diversity of agency. Potentially, this positive effect occurs because a social hierarchy is more likely to fall into place with diverse agency levels in the group, where agentic individuals naturally take on leading roles, while others are comfortable as followers. Accordingly, people might be less prone to engage in conflicts to fight for these positions (cf. Van Vugt, 2006). Additionally, it seems that leaderless groups—those without any agentic individuals driving decision making—experience the most conflict when given a task to complete. That is, a certain level of agency in a few members (i.e., having a group leader) might be good for a group to function smoothly without conflicts. Future research could explore this in an experimental setting by deliberately pairing people into groups with varying agency compositions to observe where leaders emerge naturally and how this relates to the group's conflict behaviors.

Our exploratory analysis uncovered two interesting additional insights that suggest potential directions for future studies. First, while we did not find robust actor-group fit or similarity effects of communion across tasks in our main analyses—except for selected effects present in specific tasks at the person level—differential effects of communion similarity emerged depending on the average level of communion in the group. Specifically, people seemed to engage in less task-related behaviors (cooperative, communicative, coordinated) when they were surrounded by group members who shared low levels of communion. This observation aligns with dyadic research (Arellano-Véliz et al., 2024; Cuperman & Ickes, 2009) and reinforces the idea that the benefits of communion similarly depend on its level in the group. Thus, groups including mostly highly communal members likely have an engaging atmosphere that promotes task participation, whereas groups dominated by uncaring, disagreeable members might hinder individual participation in the group.

Second, while people's liking and metaliking were unaffected by actor-group fit, the additional GAPIM-D analyses revealed a significant actor-partner fit at the dyadic level: Actors liked specific partners more when they had similar levels of agency. This finding resonates with previous research focusing on dyads (Cuperman & Ickes, 2009) and highlights the value of disentangling composition effects not only at the individual and group but also the dyadic level among group members.

In summary, our findings suggest that a complementary agency composition within groups can reduce individual and group conflict, and supplementary high communion among interaction partners can promote positive individual behaviors. However, our analysis of personality composition across the four different tasks also showed that composition effects emerged only in a few outcome domains, not consistently across tasks in most cases, and varied across

outcome levels. The following section addresses the broader implications of these findings.

Theoretical, Methodological, and Practical Implications for the Study of Personality Composition in Groups

Overall, our analysis revealed relatively few consistent composition effects at person and group levels. These findings echo those from numerous dyadic studies that frequently report small or inconsistent composition effects (see Back et al., 2023; Humberg et al., 2023; Joel et al., 2020; Weidmann et al., 2017) as well as the mixed results across group research (e.g., French & Kottke, 2013; Kristof-Brown et al., 2005; Prewett et al., 2018). What implications do these results have for understanding the role of personality composition at large? We discuss two theoretical perspectives to clarify this patchwork of findings: the importance of temporal resolution in studying personality composition and the role of task characteristics.

First, the limited impact of personality composition we observed in this study and previous research might mean that researchers need to shift toward a more time-sensitive approach to investigating personality composition in groups. That is, our focus on personality traits might be too distant from the situation, with similarities (or differences) between group members being not salient enough to impact people's behaviors, perceptions, or performances. Instead, the composition of momentary personality expressions—how agentic or communal individuals behave, think, and feel within a specific situation and how that relates to the personalities of group members in this moment—may be more important for person and group outcomes. Although we are not aware of any studies that investigate personality composition beyond the trait level in groups, research focusing on the principle of interpersonal complementarity in dyadic interactions illustrates that complementarity between communal and agentic behaviors of actors and their partners is observed more consistently than in trait-level assessments (Herde & Lievens, 2024; Kurzius et al., 2022). Importantly, a time-sensitive perspective requires researchers to think about appropriate timeframes to study personality expressions and composition effects during interactions (Bleckmann & Wagner, 2025). This perspective reflects the growing trend in personality and organizational research that emphasizes the importance of assessing and understanding dynamic interaction processes (Bleckmann et al., 2024; Lehmann-Willenbrock & Allen, 2018; Rau et al., 2022).

Second, our findings underscore the need for a context-sensitive approach to understanding personality composition in group interactions. For instance, results at the person level showed that complementary actor-group fit in terms of agency reduced individual conflict behavior in the survival on the moon task, but not in the poster design task. These discrepancies might be explained by different task characteristics, which likely shape the required communication and compromise during an interaction and thereby limit or expand the “room” for expressing personality tendencies that give rise to composition effects. Organizational and educational research has extensively examined how both objective task characteristics (e.g., complexity or content; Hackman, 1968; Tett et al., 2020) and perceived task characteristics (e.g., perceived task value; Wigfield & Eccles, 1992) relate to person and group outcomes. Similarly, there is a growing body of research on the role of psychological characteristics of situations (e.g., intellectual vs. duty related; Rauthmann & Sherman, 2020) in shaping personality expressions and people's experiences. To date, however, these different research traditions

remain largely disconnected, and research has yet to explore how personality composition effects might depend on certain objective or subjective tasks or situation characteristics (Kramer et al., 2014). In addition, while agency and communion are fundamental personality dimensions, composition of a broader range of personality traits may be important for understanding group dynamics in specific group interaction settings. For example, particularly in educational settings, it could be important for individual, dyadic, or group performance that group members share a similar level of conscientiousness to reflect a mutual sense of achievement orientation (e.g., Prewett et al., 2009).

What do the current results and the notion of a dynamic, context-sensitive approach to studying personality composition mean in methodological and practical terms? From a methodological standpoint, we see the GAPIM as a valuable tool for examining personality composition in groups with the separation of different components at person, dyad, and group levels. The additional dyadic-level analysis, though exploratory and limited to two outcomes, highlights that the choice of outcome level (e.g., studying liking at the dyadic level vs. aggregated across interaction partners) matters. Ideally, all levels should be considered for a comprehensive understanding of personality composition effects, though this requires greater resources (e.g., round-robin design, larger group sizes) and more complex analytical setups. Moreover, the GAPIM presents a flexible framework that can be extended in several ways. For example, the traditional APIM, which underpins the GAPIM, has been extended to include repeated measures data to assess actor-partner similarity over time (Savord et al., 2023). A similar extension of the GAPIM is conceivable for examining group composition dynamically.

From a practical viewpoint, our differentiated findings on agency and communion composition provide a basis for targeted feedback interventions in groups, which could be particularly relevant for work teams that experience high levels of individual and group conflict. That is, our research can help analyze and illustrate how the interaction styles of team members—shaped by their personalities and the overall composition of the team—can lead to conflict. Given that personality has stable but also dynamic aspects that can change over time (Baumert et al., 2017; Wagner et al., 2020), along with the understanding that the effects of personality composition may be time sensitive and context specific, we view personality group composition as something that is not fixed. Instead, composition can be adjusted and tailored to align with specific group goals, making it a promising starting point for improving social dynamics in teams, learning groups, and social circles. Finally, our findings suggest that group composition effects vary for different outcomes, which should be considered when assembling teams. In this regard, we see considerable potential for interdisciplinary cooperation between personality, organizational, and educational research to deepen our insights on personality composition and its underlying processes using diverse both laboratory and real-world group settings.

Limitations and Future Directions

We note several limitations in our research. First, while we consider the GAPIM a powerful and flexible framework, it is not the only statistical model for studying personality composition. For example, educational researchers often use contextual analysis models (Iversen, 1991; Lüdtke et al., 2008) to study composition in classrooms. These models share several key features with the GAPIM (Bonito, 2022), but there are also significant differences that may result in varying

outcomes, particularly in small groups. Therefore, more applications and comparisons of different composition models used across disciplines are needed to better understand variations in results (cf. Ostermeier et al., 2020).

Second, while we investigated personality composition as a predictor of individual and group outcomes, the research design did not enable us to model the underlying processes that relate stable personality traits like agency and communion to observable behaviors or people's perceptions in group settings. That is, we cannot draw any inferences about processes through which personality composition affects outcomes. Moving forward, dynamic research approaches utilizing repeated measures during social interactions could clarify these underlying processes.

Third, while we have discussed that some personality composition effects on behavioral outcomes occurred in some tasks but not in others, it is important to bear in mind that the behavior people displayed during the tasks may have been influenced by their previous behavior in preceding tasks (i.e., order effects). It is also conceivable that the behavioral coding of the tasks was partially confounded by differences between observers (due to limited resources, we were unable to have each task rated by a new team of observers). However, we assume that the differences in task demands were more decisive than the effects of the task order or observers. To verify this assumption, future research could use an optimized research design that counterbalances the task order across groups and implements a randomized coding scheme for observers.

Fourth, the effect sizes (indicated by R^2 values) observed in this study were relatively small, which is not uncommon in personality and social psychological research and seems consistent with prior studies examining personality composition effects (e.g., Gonzalez-Mulé et al., 2014; Körner & Altmann, 2023; Peeters et al., 2006). While small, these effects may still be meaningful, for example, by accumulating and thereby contributing to the prediction of group dynamics over time (Back & Vazire, 2015). Nonetheless, these findings also suggest that other factors likely play a significant role in shaping how individuals experience and behave within groups.

For example, while our research focused on the composition of people's self-perceived personality traits, perceived composition could also play an important role in group interactions. Perceived composition could include, for instance, how similar a person thinks their personality is to the other group members (i.e., perceived actor-group fit). Indeed, research suggests that actual and perceived personality similarity are conceptually distinct and have different implications for dyads and groups (e.g., Hughes et al., 2021; Selfhout et al., 2009; Shemla et al., 2016). While no studies have specifically examined perceived personality composition in groups to date, several empirical findings highlight its potential importance in social settings: First, group interaction research has shown that people's perceptions of others' agentic characteristics (i.e., if others score high or low on agency) directly affect their own behavior (Rau et al., 2019). Second, one study using GAPIM demonstrated that a person's general perceived similarity to other group members mediated the effects of actual gender similarity on their identification with the group (Garcia, 2017). This finding not only illuminates the potential interplay between actual and perceived group similarities but also showcases GAPIM's utility in modeling these complex links. Taken together, these insights suggest that how group members perceive one another, and how these evaluations fuel perceptions of

similarities and differences, might be important, independent predictors of people's behaviors and experiences in group interactions.

Finally, real-life small groups take many different forms, and our findings might not readily generalize to all of them. Our study examined same-gender groups and identified small but robust differences in behavior, experiences, and performance between female and male groups. This raises the question of whether our findings apply to mixed-gender groups, where gender itself is an important composition variable related to behaviors and performance measures (e.g., Bradley et al., 2021). For instance, research illustrates that gender stereotypes can affect how people evaluate others' behaviors and their reactions in work settings, with similar displays of agentic behavior leading to women being labeled as bossy while men are perceived as confident (Badura et al., 2018; Hemshorn de Sánchez et al., 2025). Further, this study focused on groups in early-acquaintance contexts. Consequently, personality composition may affect outcomes differently in groups wherein members have known each other for a longer time, as people are likely more aware of personality (dis)similarities.

Conclusion

The present research highlights distinct effects of personality composition both at the person and group level. Individuals' behaviors and perceptions during initial group interactions were mostly driven by their own personalities, and group behaviors and perceptions were linked most consistently to the average personality of the group. Regarding more complex composition effects, both people and groups experienced increased conflict that corresponded to the complementarity of their agency levels. At the same time, group performance was not explained by personality composition. Taken together, this study offers a framework for breaking down personality composition into distinct components at person and group levels, paving the way for a more comprehensive understanding of their respective roles in social contexts.

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